DATA SET

Taxi & Limousine Commission

NYC TLC Trip Records

Real New York taxi trips for the CDLE labs

Source
NYC TLC, one Parquet file per month

Data
Yellow taxis, Sep 2026: 3.9 M trip records

Lessons
Dirty data, Flex Fare, event time

Labs
6 pipeline · 7 Kafka · 8 streaming

CDLE • MEIC/MEIM • ISEL

Taxi & Limousine Commission

Contents

What the data is, what it hides, and how the labs use it

01 The source

Who publishes the trip records, and how often
Terms of use: download it yourself
The two files the labs read

02 The data

The schema: one trip table, five lookups
Codes as tables · 265 zones, two unknown
One month in numbers

03 What real data teaches

Eight kinds of dirty rows, twins included
Plausible ≠ true: Flex Fare and negative fares
Event time: a trip exists only when it ends

04 Hands-on

Where Labs 6, 7 and 8 use the data
Getting the data, with a checksum

The TLC publishes every trip, month by month

The NYC Taxi & Limousine Commission (TLC) collects the trip records from the providers in the cars

Yellow taxis
Street hails and app bookings, mostly Manhattan; meter fields, itemised fares
used in CDLE

Green taxis
Street hails outside central Manhattan; like yellow, but lpep_ times and trip_type

For-hire vehicles
Pre-booked cars from bases; few columns

High-volume FHV
App companies (Uber, Lyft, …); own, richer data set since 2019

How it is published

- One Parquet file per service and month
- A new month appears about 2 months later: September 2026 was the latest on 6 Nov 2026
- Since 2025 the yellow, green and HVFHV files carry cbd_congestion_fee

What CDLE uses

- Yellow taxis only
- September 2026: 3,876,044 trip records, 67 MB
- August 2026 only for Lab 6's optional backfill
- The taxi zone lookup (265 zones)

Source: TLC Trip Record Data page, nyc.gov/site/tlc/about/tlc-trip-record-data.page (6 Nov 2026); data dictionaries of 18 Mar 2025.

NEW TERMS

`Street hail` passenger flags a taxi down on the street
`Meter` in-car device that computes the fare
`Parquet` columnar file format for large data
`cbd_congestion_fee` Manhattan congestion toll, since 2025
`Backfill` re-running a pipeline for past dates
`Taxi zone` one of the TLC's 265 NYC areas

Terms of use: download it, do not redistribute

Open data with no licence on the page: as with the TPC-DS kit, each student gets it from the source

What the TLC says

- The records come from the technology providers; the TLC did not create them
- The TLC makes no representations as to their accuracy
- The page states no licence; NYC.gov terms of use reserve the city's rights

The CDLE rule

- Each student downloads the files from the TLC
- We do not put them on Moodle or in Git, and you do not either
- Aggregated results in your reports are fine: normal academic use

In practice

Download from the TLC | **Check the SHA-256** | **Same file for everyone** | **No copies on Moodle or Git** | **Say if it changed**

Sources: TLC Trip Record Data page and Trip Record User Guide (nyc.gov), 6 Nov 2026.

NEW TERMS

`Open data` data published free for anyone to use
`TPC-DS` standard benchmark with a data-generator kit
`Technology provider` runs the in-car meter system
`SHA-256` checksum that proves a file is unchanged

Two files are all the labs need

One month of yellow taxi trips in Parquet, plus a 12 KB lookup table that turns zone IDs into names

File	Size	Rows	Columns	Used for
yellow_tripdata_2026-09.parquet	67 MB	3,876,044	21	Labs 6, 7 and 8
yellow_tripdata_2026-08.parquet	59 MB	3,336,716	21	Lab 6, optional backfill
taxi_zone_lookup.csv	12 KB	265	4	zone names (all labs)

Where they live

- **Host:** d37ci6vzurychx.cloudfront.net
- **Trips:** /trip-data/yellow_tripdata_YYYY-MM.parquet
- **Zones:** /misc/taxi_zone_lookup.csv
- Linked from the TLC Trip Record Data page

Inside the Parquet

- 4 row groups of about 1 M rows each, so Spark can split the file
- One day of trips is about 2–3 MB: small enough for a laptop, big enough to be real
- Times are local New York time, with no time zone

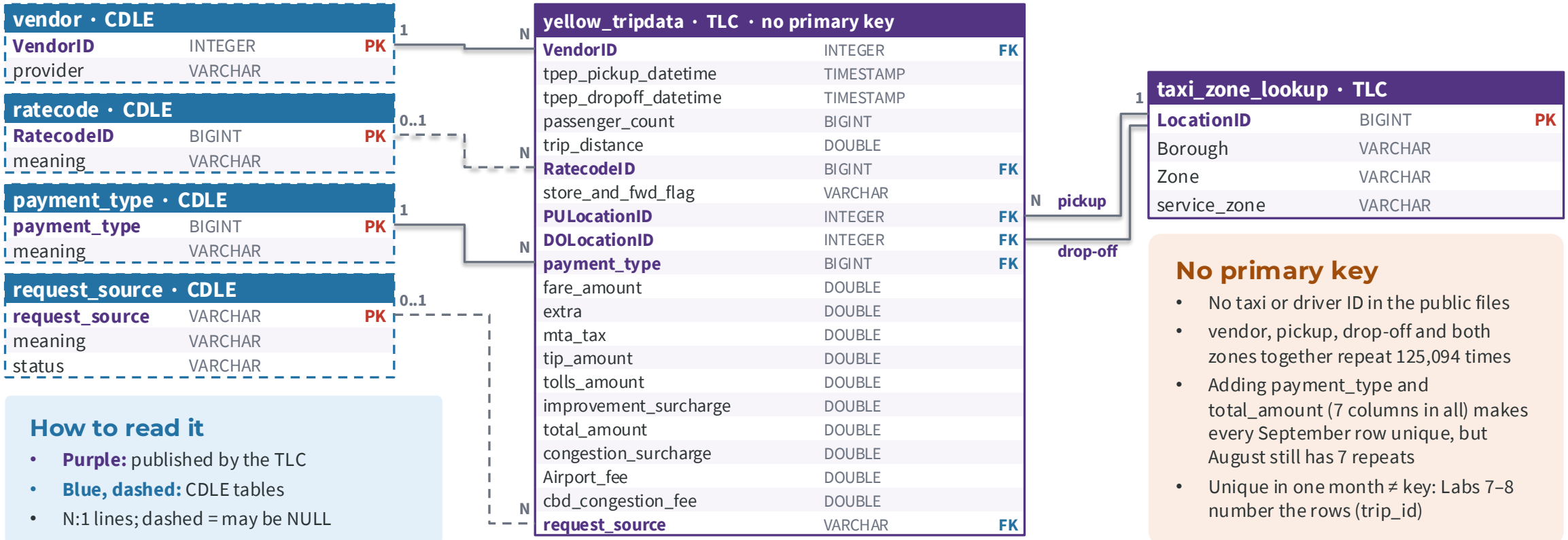
Source: yellow_tripdata_2026-09.parquet (TLC, downloaded 6 Nov 2026), measured with DuckDB 1.4.5.

NEW TERMS

- **`Lookup table`** small table that maps codes to names
- **`CloudFront`** Amazon service that hosts the TLC files
- **`Row group`** horizontal chunk of rows in a Parquet file
- **`DuckDB`** in-process SQL engine for local files

The schema: one trip table, five lookups

Trips join the zones twice (pickup, drop-off) and four code tables that we (CDLE) built from the TLC dictionaries



How to read it

- **Purple:** published by the TLC
- **Blue, dashed:** CDLE tables
- N:1 lines; dashed = may be NULL

No primary key

- No taxi or driver ID in the public files
- vendor, pickup, drop-off and both zones together repeat 125,094 times
- Adding payment_type and total_amount (7 columns in all) makes every September row unique, but August still has 7 repeats
- Unique in one month ≠ key: Labs 7–8 number the rows (trip_id)

Sources: TLC data dictionaries (18 Mar 2025); Parquet types of yellow_tripdata_2026-09.parquet (TLC, downloaded 6 Nov 2026), measured with DuckDB 1.4.5.

NEW TERMS

- **PK / FK** primary key (row ID) / foreign key (link)
- **tpep_** prefix from TPEP, the yellow-taxi meter system
- **Flex Fare** price fixed upfront when booked by app
- **Bronze/silver/gold** raw, clean, curated data layers

Codes: four small tables we (CDLE) built

The TLC publishes the codes only in a PDF; as tables, a pipeline can join them and check them

`vendor` · CDLE table

VendorID	Provider	Rows, Sep
1	Creative Mobile Tech.	694,912
2	Curb Mobility	3,128,003
6	Myle Technologies	5,236
7	Helix	47,893

`payment_type` · CDLE table

payment_type	Meaning	Rows, Sep
0	Flex Fare trip	1,190,413
1	Credit card	2,332,127
2	Cash	321,102
3	No charge	10,509
4	Dispute	21,891
5	Unknown	2
6	Voided trip	0

`ratecode` · CDLE table

RatecodeID	Meaning	Rows, Sep
1	Standard rate	2,448,599
2	JFK	91,759
3	Newark	9,897
4	Nassau/Westchester	7,707
5	Negotiated fare	30,962
6	Group ride	0
99	Null/unknown	96,707
NULL	(Flex Fare trips)	1,190,413

Why tables

- A join gives names in the gold tables
- An anti-join finds codes the dictionary does not know
- A code nobody documents stays “undocumented”: no guessing

`request_source` · CDLE table · HV = TLC licence numbers (HVFHV dictionary)

request_source	HV0003	HV0005	A	EH0004	CC	EH0010	NULL
Meaning	Uber	Lyft		*undocumented*			*(not Flex Fare)*
Rows, Sep	709,010	408,034	63,426	8,733	580	309	2,685,952

Sources: TLC yellow taxi data dictionary (18 Mar 2025); rows in yellow_tripdata_2026-09.parquet (TLC, downloaded 6 Nov 2026), measured with DuckDB 1.4.5.

NEW TERMS

`Pipeline` chain of dependent jobs run on a schedule

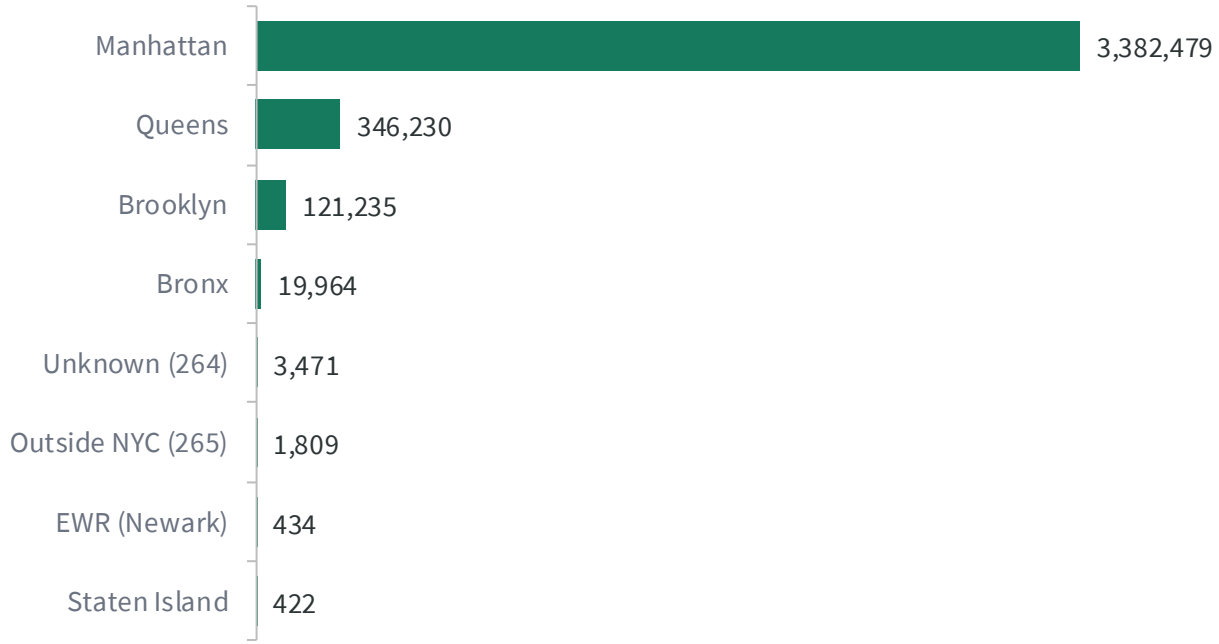
`Data dictionary` defines every column and code

`Anti-join` keeps rows with no match in the other table

Zones: 265 IDs, two of them mean "unknown"

PULocationID and DOLocationID join to taxi_zone_lookup.csv: LocationID → Borough, Zone, service_zone

Pickups by borough, September 2026



265 zone IDs

- 69 Manhattan · 69 Queens · 61 Brooklyn · 43 Bronx · 20 Staten Island · 1 EWR
- 264 = Unknown, 265 = Outside of NYC (its Borough is "N/A")

25,940 trips (0.7%) touch 264 or 265

at pickup or drop-off: no real borough or zone

87% of pickups are in Manhattan

Top: UES South 173 k · UES North 155 k · JFK (Queens) 152 k

Source: yellow_tripdata_2026-09.parquet (TLC, downloaded 6 Nov 2026), measured with DuckDB 1.4.5; taxi_zone_lookup.csv. Borough counts are pickups (PULocationID).

NEW TERMS

- **PU / DO** pickup / drop-off, as in PULocationID
- **Borough** one of NYC's five boroughs
- **EWR** Newark airport, New Jersey; its own zone
- **service_zone** zone class: Yellow Zone, Boro Zone, airport

One month in numbers: 3.9 million records

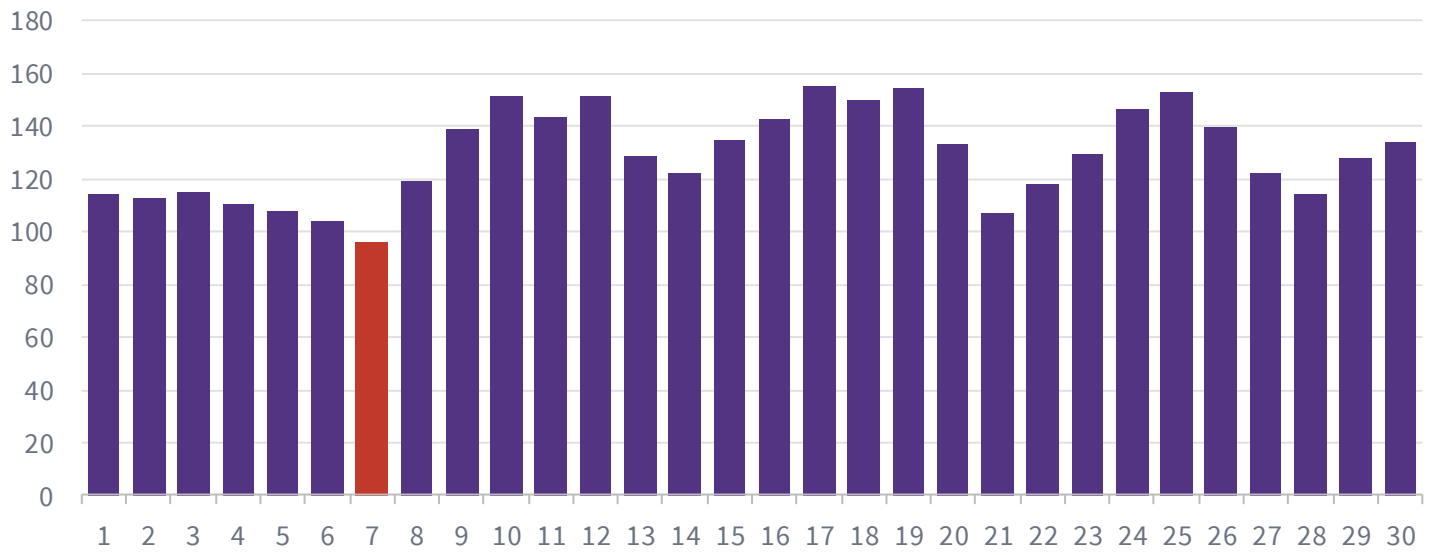
3.88 M
trip records in **September 2026**

129 k
trips on a median day (96–155 k)

15.4 min
median trip · 1.96 miles

124.6 M$
total_amount for the month

Trips per day (thousands), by pickup date



How trips were paid

- **Card** 2.33 M (60%)
- **Flex Fare** 1.19 M (31%)
- **Cash** 0.32 M (8%)
- **Dispute / no charge** 32 k

Read the chart

- Red bar: 7 Sep, Labor Day, the quietest day
- Weekly rhythm: Thursday to Saturday peak
- Mondays (7, 14, 21, 28): the low of each week

Source: yellow_tripdata_2026-09.parquet (TLC, downloaded 6 Nov 2026), measured with DuckDB 1.4.5. Records, not trips: 112,068 trips appear twice (slide 10).

NEW TERMS

`Median` middle value: half above, half below
`Labor Day` US holiday, first Monday of September
`Record` one row; a trip can have two
`Dispute` payment code: the rider contested the fare

Real data is dirty: eight problems in one month

Every problem below is in the September file: a pipeline must decide what to do with each one

Problem	Rows	%
trip_distance = 0	147,158	3.8
Same trip twice (VendorID 1: Flex Fare + meter)	112,068	2.9
RatecodeID = 99 (unknown)	96,707	2.5
Duration 0 (pickup = drop-off)	48,202	1.2
Zone 264/265 at pickup or drop-off	25,940	0.7
Negative fare_amount	14,190	0.4
Trips of 3 h or more (18 of 24 h or more)	852	0.02
Pickup outside September (back to 2009)	22	0.00

Quarantine and count

Do not delete silently: move bad rows aside, count them, look at them

Twins, not copies

No two rows are identical, but every VendorID 1 Flex Fare trip has a meter twin: same times, zones, distance, fare. 2.8–3.3 M$ counted twice

Not on the list

1.19 M NULL passenger_count rows: Flex Fare, not errors

Source: yellow_tripdata_2026-09.parquet (TLC, downloaded 6 Nov 2026), measured with DuckDB 1.4.5. A row can have several problems; percentages of all 3,876,044 rows.

NEW TERMS

- `Dirty data` rows that are wrong, missing or odd
- `Quarantine` set bad rows aside instead of deleting
- `Twin trip` Flex Fare row with an identical meter row

Plausible ≠ true: two rules that look right

Both rules are wrong for this data: check every rule against the rows before you trust it

"passenger_count is not null"

- **1,190,413 rows fail (30.7%).** All of them have payment_type 0: Flex Fare, an upfront price from an app
- Flex Fare trips also leave RatecodeID and store_and_fwd_flag NULL
- 94% of them were booked through Uber (709 k) or Lyft (408 k)
- **The rule would quarantine a third of the valid trips.** Right rule: only when payment_type <> 0

"fare_amount >= 0"

- **14,190 rows fail (0.4%).** 12,500 of them have a positive twin: same vendor, times and zones
- They are voids and disputes: payment_type 4 (dispute) in 8,671 of the 14,190
- The pair cancels out: + X and − X add up to zero
- **Dropping only the negative half inflates the day's revenue.** Keep both, or neither

Where you will meet this

Lab 6 quality gate | **AI-drafted expectations** | **Audit: keep, adjust, reject** | **Deck 6: AI drafts, you audit**

Sources: TLC data dictionary; yellow_tripdata_2026-09.parquet (TLC, downloaded 6 Nov 2026), measured with DuckDB 1.4.5.

NEW TERMS

- `Void` cancelled charge, reversed by a negative row
- `Quality gate` check that stops bad data moving on
- `Expectation` a testable rule the data must meet

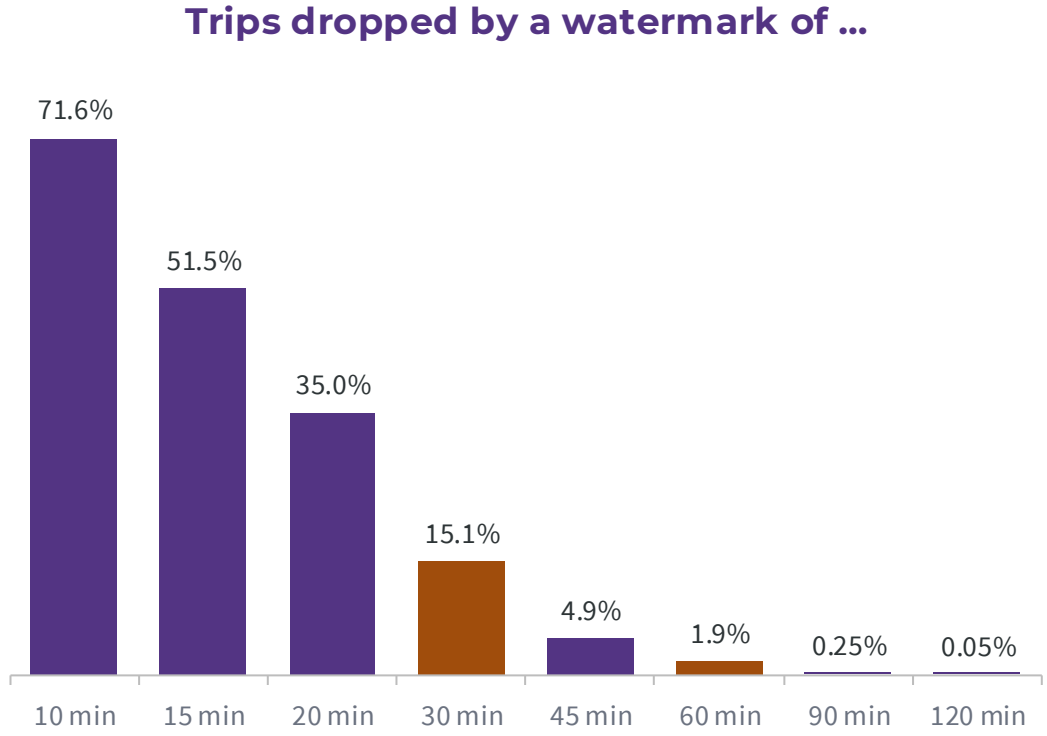
A trip exists only when it ends: event time

The record is written at drop-off, but we count by pickup time: delay = length of the trip



Why it matters for streaming

- A 5-minute window by pickup keeps getting trips after it ends
- A watermark says how long to wait; later trips are dropped
- **store_and_fwd_flag = Y:** 3,881 trips held in the car: late arrivals
- 5-minute windows on 15 Sep: median 518 trips, peak 928
- Spark moves the watermark once per micro-batch: Lab 8 measures what it really drops



Source: yellow_tripdata_2026-09.parquet (TLC, downloaded 6 Nov 2026), DuckDB 1.4.5. Chart: share of trips longer than W (delay = duration), whole month.

NEW TERMS

- `Event time` when it happened, not when it was recorded
- `Window` time bucket that groups events, e.g. 5 min
- `Watermark` how long a stream waits for late events
- `store_and_fwd_flag` Y = trip stored, sent later

PART 2 · Information about the practice

what you will do with the data

Labs 6–8: pipeline, Kafka, streaming

Where each lab uses the data

The same September file in all three labs; Labs 7 and 8 read it directly, not your Lab 6 results

Lab	What it reads	What you do with it	Part 1
6 · Orchestrated pipeline	September, one day per run (logical date); August only for the optional backfill	ingest → clean → validate → publish in Airflow; run twice, kill, backfill; an AI-drafted quality gate you check	slides 10–11
7 · Kafka	September: the 134,352 trips that end on 15 Sep, the Lab 6 day	replay them into a topic in drop-off order, keyed by pickup zone; crash a consumer; evolve the schema	slide 12
8 · Stream processing	the Lab 7 day again (134,352 trips); at home, the week of 14–20 Sep (991,650)	5-min windows by pickup time in Spark; bet on a watermark and count what it drops; kill the job and resume	slide 12

Lab 6: a lake per day

bronze → silver → gold, one folder per day (day=2026-09-15), overwritten, never appended

Independent labs

Labs 7 and 8 need only the September file: a Lab 6 that went wrong does not block them

What you will meet

Flex Fare NULLs and twin trips (Lab 6); duplicates after a crash (Lab 7); late events and watermarks (Lab 8)

Counts from yellow_tripdata_2026-09.parquet, by drop-off day: the order in which the trips reach the topic.

NEW TERMS

- **`Airflow`** open-source tool that schedules pipeline tasks
- **`Logical date`** the day a run processes, not today
- **`Data lake`** plain files in folders, read by many engines
- **`Kafka`** distributed log that streams events
- **`Topic`** a named stream of messages in Kafka
- **`Replay`** re-send stored records as if live

Getting the data: one command, checked

Run `get_data.py` (Labs 6–8 starters) before the class with `$PY`, your Lab 1 Python: no container needed

```
# September 2026 and the zones (about 67 MB)
cd ~/tpcds/lab6      # or lab7, lab8: the same script
$PY get_data.py

# optional, Lab 6 only: August 2026, for the backfill (59 MB)
$PY get_data.py --month 2026-08

# what you should see (a second run downloads nothing)
ok   yellow_tripdata_2026-09.parquet  (67.4 MB, …, checksum right)
ok   taxi_zone_lookup.csv  (0.0 MB, …, checksum right)
files in ~/tpcds/nyc_taxi
```

What it does

- Downloads from the TLC to ~/tpcds/nyc_taxi/ (the lab's containers see it as /data/raw)
- Checks each file against the SHA-256 of 6 Nov 2026, written in the script
- Skips a file that is already there with the right checksum

If it says DIFFERENT

The TLC changed the file: do not fix it, tell the teacher

Keep the file

Labs 7 and 8 read the same September file

Do not redistribute

No copies on Moodle or Git (slide 4)

Source: get_data.py in the Lab 6–8 starters (the same in all three); output from a run on 6 Nov 2026 (the times of a first download vary).

NEW TERMS

`$PY` your Lab 1 Python command

`Lab starter` files we give you to start a lab